Binary Trees

specific kind of tree with every node having 2 children

just 2 children.

left right

→ makes the structure simpler than a generic tree
→ extremely powerful for various computational purposes

Real life example.

→ Family tree with 2 parent
→ Yes/No Decision making
→ Tournament Bracket

Industry example

→ Binary DT in AI
→ Database Searching (mainly BST)
→ Binary Trees in compilers

Binary Tree node

Now no list is needed, just a left & right will do

2 subtree



Print function

→ now loose code would be required

also now no loop now & only left, right would do

inorder preorder post order

No duplicate

Tree from inorder and preorder

not to make mistake of in S[0]

Diameter of a binary tree

max^m distance b/w 2 nodes

→ Check balanced → Sum of nodes → find min max

→ Tree from inorder and post order

→ Tree from inorder and pre-order